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Optical | North America

ECOC First Take: Full Steam Ahead

Versus OFC, consensus at ECOC was that NPO is coming in a 2028 time frame in parts of scale-up networks. Architectures are likely to evolve dramatically as technology advances, but optical content to continue to increase. Supply chain largely sold out; InP substrate seemingly the biggest bottleneck.

Key Takeaways

- Consistent with OFC, optical vendors are seeing continued strength in demand; not a surprise to investors at this point in the AI capex cycle.
- Incremental to the conversation from OFC, there is more confidence in NPO adoption, as Open CPX MSA introduces an ecosystem around solutions.
- OCS to still see growth, but Google to by far remain the largest implementation.
- CSCO and NOK being opportunistic with supply, but do not see major share shifts, with CIEN holding majority line system share.
- Prefer KEYS, but with early NPO likely to use external lasers (which then need to be higher power), see LITE/COHR continuing to have most attention.

This note is a first take summary of meetings we hosted at ECOC in Spain on 9/21 and 9/22. We will publish more detailed company specific takeaways from our meetings with CSCO, CIEN, COHR, KEYS, NOK (covered by Terence Tsui), TSEM, Lightmatter in the coming days.

Sunny skies in Spain as sun shines down on optical. ECOC has traditionally been a smaller, more technical trade show to OFC. However, with the optical trade now well discovered as content in AI networks grows, this event has garnered much more investor attention. **Themes brought up at the show are no surprise: optical content is growing faster than suppliers can produce it, many vendors are sold out for 12-18 months, and the exact form that CPO/NPO will take is still up for grabs.** However, versus 6 months ago, there was more recognition that NPO is going to happen starting in 2028, aided by the Open CPX MSA, which is seeking to build common parameters for optical engines that will allow a greater ecosystem to be built, making hyperscalers less concerned about adopting it than when NVDA's solution was the only option (NVDA is covered by Joe Moore). This isn't to minimize that there are still tremendous scaling / production / implementation challenges ahead in the coming years. It is just to say that there are enough suppliers working towards an open solution, and a growing need for optics and electronics to be closer to each other in scale-up, that a first generation of product is going to at least be trialed in 2028. There was also consensus that the architecture the first generation of product takes is unlikely to be the final format, as laser types, where those lasers are located, etc., are all subject to change in coming generations. This also means

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competitive positioning today isn't a lock for the long term.

Common themes from meetings:

- **End goal: use optics to reduce power, cost, and latency, while improving bandwidth density, capacity, and the reliability of high-speed AI networks.**

The above line is the stated goal of the Open CPX MSA, highlighting that while much is made about the large, growing opportunity for optics vendors, the end goal of hyperscalers is to find the most power and cost efficient solutions (e.g. the reason we have seen ZR pluggables proliferate, why network builders keep trying to make silicon photonics (CW) lasers have greater applications, the adoption of XPO). While we are not calling for margins to peak anytime soon, we are mindful that as opportunity multiplies, the form solutions take today may not be the form they always take (e.g. VCSELs could see a resurgence in short reach applications). Overall, this remains a positive for nearly everyone in the space, but we still prefer **KEYS** given less dependent on end destination, just that solutions are being trialed.

- **Everyone working towards an NPO solution.** While we still had one meeting where CPO/NPO was described as "inevitable, but not imminent", the general consensus is that hyperscalers are going to move towards NPO in coming years. This is different than at OFC, where there was a resistance to NVDA's CPO solution given concerns around vendor lock-in / closed ecosystem. However, **COHR**, with their PhotonLink announcement, addressed an upcoming challenge in that hyperscalers may want to move to open solutions, but don't necessarily have all the knowledge in house to assemble a solution from 6+ different vendors. PhotonLink gives customers an end-to-end solution for CPO/NPO, proving to them it can be done, and allowing customers to pick the pieces of the solution they want. We think COHR's broad based portfolio is particularly suited for this, with the company noting they have an LTA with a customer for an NPO solution that should begin ramping next year. While we did not meet with **LITE** at the show, we would note that they have talked about expecting an LTA announcement in coming months for NPO. While COHR/LITE are loudest, we would continue to note that many vendors are working on NPO solutions.
- **Mostly sold out, but not uniformly so.** In general, most component / system suppliers were sold out for the next 12-18 months, with InP substrate being the gating item (e.g. AXTI, Sumitomo). To the extent that more substrate comes available, or yields improve, there can be upside, but conditions remain tight amongst EMLs, CW lasers, pump lasers, and gold boxes, just to name the major categories. We would note that **CSCO** and **NOK** did talk to still having some availability, which we think allows them to be strategic in the near term around supply releases, but do not expect any meaningful share shifts.
- **OCS is coming, but likely smaller use cases with subsequent customers.** Google's use of OCS is well reported, with production ramp moving to LITE in coming years. Because of Google's success with the product, nearly every other hyperscaler / neocloud is trialing OCS. From what we heard at the show, there will be many more customers for OCS in the coming years. However, those use cases are likely to be smaller in scale than Google's

implementation (with one customer apparently canceling an implementation after fully qualifying a solution). We still see it as a margin accretive opportunity for LITE and COHR, but would expect LITE to accrue most of the value in the near term as supplying into the largest implementation.

Prefer those that benefit no matter the path. We have been very vocal in liking **KEYS**, seeing it as able to benefit from the growth in optics, no matter the implementation. We would tend to put **GLW** in that same boat as a beneficiary, no matter the implementation. However, we would still note, **LITE / COHR** are likely to remain fairly sold out in the foreseeable future as InP lasers remain in short supply, and first generation NPO/CPO products are likely to all use external lasers, meaning high powered lasers will see dramatic increases in demand in coming years. **CIEN** has multiple paths to success, but is dependent on InP supply chain loosening more in coming years to be able to materially outperform the target laid out (which is still not fully captured in price). We did not pick up on any reason for **CSCO** weakness today in meetings, as demand conditions remain strong, and supply chain being more in their control gives them continued opportunities to grow their revenue base from cloud customers.

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Cisco Systems Inc (CSCO.O)	O (04/08/2024)	\$106.44
Coherent Corp (COHR.N)	E (12/13/2023)	\$310.39
Corning Inc (GLW.N)	E (06/13/2024)	\$159.69
F5 Inc (FFIV.O)	E (04/12/2022)	\$453.33
Keysight Technologies Inc (KEYS.N)	O (07/13/2026)	\$347.32
Lumentum Holdings Inc (LITE.O)	E (05/12/2021)	\$945.67
Motorola Solutions Inc (MSI.N)	O (12/17/2025)	\$456.70
Zebra Technologies Corporation (ZBRA.O)	E (12/02/2024)	\$361.61

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.